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Practice Problem-3: 26 Sept. 2023
Subject: CSE 1202: Digital Logic Design Lab

INSTRUCTIONS:

- a. Use + line of the breadboard for shorting **5V** and – line for shorting **GND**.
- b. Use a minimal number of **ICs** and **wires** for the hardware implementation.
- c. **Follow the orientation of the higher significant bit to lower significant bit from left to right or up to down.**

Question – 1

Marks

Consider that, the following gates are available:

- Not gate
- 2 input OR gate
- 2 input AND gate

Now simplify the following boolean expression so that the number of required gates are minimum. **X**

$$F(A, B, C, D) = \bar{A}\bar{B}\bar{C} + \bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}$$

Mention the number of each gate properly required.

Question – 2

Dhrubo lives in Kishoreganj, where it's tricky to guess if it will rain. He often doesn't know if he should take an umbrella when he goes out. So, he wants your guidance in this regard. You decided to check how the weather works in Kishoreganj and found that three parameters of weather matter: the condition of sky, temperature and humidity.

In Kishoreganj, if the sky is cloudy, the temperature is low, and the humidity is high, then it's definitely going to rain. In that case, you should tell him to carry an umbrella. But if the sky is cloudy and the temperature is high, it's a bit confusing. It might rain, or it might not. So, in these cases, you can suggest as per your wish whether he should carry an umbrella or not. It is completely your choice. But for other situations, it doesn't rain in Kishoreganj, so you should advise him not to carry an umbrella.

Now, you need to make a simple decision, represented by a boolean expression F about whether Dhrubo should carry an umbrella or not, based on the parameters of weather. The representation of the states of the parameters of weather and F is stated in the following table. NB: Your suggestions should be in such a way that F requires a minimum number of gates to be implemented. **X**

	1	0
Sky (A)	Cloudy	Not cloudy
Temperature (B)	High	Low
Humidity (C)	High	Low
Carry umbrella (F)	Yes	No

Question – 3

Simplify the following boolean function:

- $F(A, B, C, D) = \sum m(0, 2, 3, 5, 6, 8, 9) + d(10, 11, 12, 13, 14, 15)$
- $F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$

Mention the number of each gate properly required. **X**